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标题	Secondary Phosphine Oxide-Stabilized Pd Nanoparticles: Ligand-Controlled Chemoselective Hydrogenation of Nitro Compounds toward Amines in Water
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摘要（原文节选）

A novel approach to the catalytic synthesis of aromatic amines using secondary phosphine oxides (SPO) supported on palladium nanoparticles is described. In this method, the ligand modulates the chemoselectivity of the target product in aqueous conditions. The catalyst demonstrated exceptional synergistic effects with water in the selective catalytic hydrogenation of nitro compounds and nitrogen heterocycles. The presence of water and ligand is expected to enhance substrate uptake and facilitate the synergistic acti ⋯

数据表 · 1（13 行，含数值行 13 行）

Palladium Nanoparticle-Catalyzed Hydrogenation of Nitro Compound (1a): Conditions Screening

entry	catalyst	solvent	time (h)	yield (%)
1	SPO-Pd-NPs	DCM	24	11%
2	SPO-Pd-NPs	Dioxane	24	trace
3	SPO-Pd-NPs	Toluene	24	0%
4	SPO-Pd-NPs	DMSO	24	<5%
5	SPO-Pd-NPs	CH3OH	24	18%
6	SPO-Pd-NPs	H2O	24	96%
7	SPO-Pd-NPs	H2O	24	95%
8	SPO-Pd-NPs	H2O	12	88%
9	/	H2O	24	0%
10	SPO-Pd-NPs	Solvent free	24	40%
11	Pd/C	H2O	24	37%
12	Pd (PPh3)4	H2O	24	21%
13	PPh3-Pd-NPs	H2O	24	19%

脚注（逐字）：aReaction conditions: Nitrobenzene (1a, 0.5 mmol), solvent (1 mL), SPO-Pd-NPs (1 mol %, 0.8 mg, based on Pd), H2 balloon, room temperature.bGC yield.c80 ° C.

正文对应句（逐字）：Nitrobenzene 1a was initially selected as the model substrate. Solvent choice was critical for the synthesis of the target anilines, as detailed in Table . Successful trials yielded aniline 2a at 11% through selective catalytic hydrogenation (entry 1). Encouragingly, the yield of target product 2a i

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